

Low-Frequency Autoregressive Model with Markov-Switching

Feng Chang

Abstract

Existing hidden Markov regime-switching models based on Hamilton (1989) characterize a time series process as a sequence of stochastic, segmented fix trends with some short-term dynamics in mean or conditional variance. To capture the time-varying trend and conditional volatility of economic time series in low-frequency, this paper extends the Hamilton model to the regime duration domain by assuming that the current process depends on the current regime duration (number of periods that the process has been in the current state) and few previous regime durations. Deviation from the long-term trend occurred over the previous durations is assumed to have an impact on the trend of the current process as well as the entire current regime duration. So this model generates interactions not only between the high-frequency current trend and low-frequency variables which aggregate the information over each of the past regime durations, but also between the current regime duration and the previous regime durations. Similarly, I have also incorporated a low-frequency autoregressive conditional variance process in the model. By transforming the state-space representation to duration-space representation in describing the Markov process and truncating the path-dependency by a maximum allowable total duration, I am able to alleviate the path-dependency problem involved and keep estimation of the model tractable. The maximum likelihood estimation, the basic filtering presented in this paper produce reasonable results when I apply the model to the quarterly data of foreign exchange rates for British Pound, French Franc, German Mark, and Japanese Yen over 1973-1994. Significant low-frequency mean and conditional variance autoregressive processes in the data have been detected by the model.

This article is from the third chapter of my Ph.D. dissertation from the University of California, Irvine. The author is grateful to Nai-fu Chen, Charles Cuny for their helpful comments.

Low-Frequency Autoregressive Model with Markov-Switching

1. Introduction

Business cycle plays a very important role in explaining the time series properties of GDP and asset returns. Cochrane (1988) and Fama and French (1988) show that the conventional econometric procedures based on low-order short-horizon parsimonious models overstate the random walk component in GDP or stock prices. By investigating the long horizon properties of the data, they find that both series contain a large temporary component which suggests a strong long-term mean-reversion. Stock and Watson (1988) find that variations in growth trends constitute a quantitatively large part of the movement in real per capita GDP and the changes in this stochastic trend are related to the business cycle movements. Chen, Roll, and Ross (1986), Fama and French (1989), and Fama (1990) show that stock returns are related to business conditions. Chen (1991) further shows that the state variables, which describe the relative phases in a business cycle, can be used to explain the intertemporal risk and return of assets. Daniel and Torous (1991) illustrate a strong coherence between stock returns and industrial production growth rates in the business cycle frequency based on time domain correlation smoothing. All the above researches suggest that using the right econometric tool is crucial to analyze the low-frequency properties of economic time series.

Three problems can possibly arise when one uses the conventional econometric methods to analyze the low-frequency dynamics of a time series. First, the time series may not be covariance stationary if business cycle follows a discrete structural shift process that drives the movements of the observed economic time series. Thus, applying the conventional econometric tools requires certain precaution because it assumes a single structure for the conditional mean

and variance. In detecting the low-frequency properties of a time series, the conventional econometric methods will encounter another problem: if using high-frequency data, one will have enough observation number and not be able to detect the business cycle relationship; if using low-frequency data, one will be able to detect the business cycle relationship but not have enough data points to obtain statistical power. Finally another problem involved is how to aggregate short-term information into long-term business cycle information. Most of the previous researches takes fix interval time-aggregation approach on data which exhibit time-varying business durations. This approach has its pitfall because one might over or under aggregate information such that the true business cycle relationship may not be revealed very well.

Hamilton (1989) adopts a Markov 2-state switching model to describe the GDP growth assuming two different fix growth trends. Others use the regime-switching model to analyze important economic variables such as interest rates [Gary(1996)], stock returns [Bekaert and Harvey(1995)], and foreign exchange rates [Engel and Hamilton(1990)]. The coefficients in the regime-switching model are different in each regime to account for possibility of multiple mechanisms over the sample period. The regimes, however, are unobservable and probabilistic statements can be made about the relative likelihood of their occurrence at time t given the information set. The regime-switching model is more complex, flexible in structure, and therefore is able to cope with the problems just addressed under a single, unified framework.

Despite of its success in dealing with issues related to nonstationarity, statistical power, and time-aggregation, the Hamilton's model does not allow too much low-frequency dynamics. In the existing regime-switching literature, the trend and variance are usually assumed to be constant within each regime and there is no time-varying trend or variance dynamics across

regime durations since the autoregressive process is only used to describe the high-frequency dynamics in the form of error correction around the long-term trend or the high-frequency dynamics in conditional volatility. This specification may appear to be adequate to describe time series such as U.S. GDP with strong constant trend (or variance) for the same regime (expansion or recession) over different time periods. However, the growth trend for the most financial time series such as stock prices, interest rates, and foreign exchange rates does not appear to be constant for the same regime (e.g. bull or bear markets) over different time periods, though they may appear to be constant within any one duration. Clearly, we need models which can incorporate low-frequency time-varying trend and variance (not within regime duration, but across regime durations).

In this paper I will introduce a model which allows low-frequency autoregressive process across regime durations and hence is able to generate time-varying segmented trends and volatilities. The current process is assumed to depend not only on the inferred current state and the number of periods the process has been in that state, but also on all the states of the previous two regime durations (so it allows path-dependency for at least a full-cycle in case of two distinct states). We include only the lagged low-frequency variables (variables which aggregate information for each duration in the past) in the model to explain the high-frequency data as well as to describe the low-frequency autoregressive process in the mean, variance, and Markovian transition probability. The estimation of the general path-dependent model could be very computational intensive because the entire history of the state-switching process is involved. However since we only need to consider the most recent three regime durations, we are able to achieve significant computation reduction by transforming the state-space representation into duration-space representation in describing the Markov process.

The remainder of this paper is organized as follows. Section 2 discusses the related existing regime-switching models and then introduces the low-frequency autoregressive model with regime-switching. Section 3 presents the maximum likelihood estimation procedures as well as a method to reduce the computational intensity. Section 4 applies the model to analyze the foreign exchange rates and reports the empirical estimates of the model. Section 5 concludes the paper.

2. A Low-Frequency Path-Dependent Model with Markov-Switching

In this section, I will first layout the general framework of path-dependent model with Markov-switching. Then I will discuss some existing regime-switching models in the context of path-dependency. Finally I will present a low-frequency path-dependent model with autoregressive process across regime durations.

2.1 General Framework of Path-Dependent Model with Markov-Switching

Let s_t be an unobservable random variable that takes only integer values $\{1, 2, \dots, J\}$ and y_t be an observable time series. Also denote a segmented set of any time series x as $X_{t_1}^{t_2} = (x_{t_1}, x_{t_1+1}, \dots, x_{t_2})$ so that $S_1^t = (s_1, s_2, \dots, s_t)$ and $Y_1^{t-1} = (y_1, y_2, \dots, y_{t-1})$. The general path-dependent model with Markov-switching takes the following form:

$$y_t = \mu(S_1^t, Y_1^{t-1}) + \sigma(S_1^t, Y_1^{t-1})\xi_t, \quad \text{where } \xi_t \sim \text{i.i.d. } N(0, 1). \quad (1)$$

Both the mean and variance components of (1) are functions of the entire history of the markov process and the time series Y_1^{t-1} . s_t is presumed to follow a high order J -state Markov chain with transition probabilities $P\{s_t | S_1^{t-1}, Y_1^{t-1}\}$ and $\sum_{s_t=1}^J P\{s_t | S_1^{t-1}, Y_1^{t-1}\} = 1$.

2.2 Existing Path-Dependent Models with Markov-Switching

The major concern about the path-dependent models with regime-switching is their computational intensity. If a process depends on the long past history of state variables, it is essentially computationally intractable and hence impossible to estimate. Existing regime-switching models have effectively kept the computation intensity low mainly by first assuming a low-order path-dependency and then transforming the low-order Markov process into a first-order Markov process.

Hamilton (1989) analyzes the U.S. real GNP growth based on the following model:

$$y_t - \beta_{s_t} = \phi_1(y_{t-1} - \beta_{s_{t-1}}) + \phi_2(y_{t-2} - \beta_{s_{t-2}}) + \phi_3(y_{t-3} - \beta_{s_{t-3}}) + \phi_4(y_{t-4} - \beta_{s_{t-4}}) + \sigma \xi_t;$$

where $\xi_t \sim N(0,1)$ and s_t is presumed to follow a first order two-state Markov chain with constant transition probabilities $p_{ij} = P\{s_t = j \mid s_{t-1} = i, s_{t-2} = k, \dots\} = P\{s_t = j \mid s_{t-1} = i\}$. Intuitively, this model describes a mean switching process with high-frequency autoregressive error corrections around the long-term mean. This model allows certain degree of path-dependent in the mean process, in which y_t depends on 4 lagged high-frequency past states $s_t, \dots, s_{t-4}$. He defines a new state variable s_t^* that characterizes all the possible combination of $s_t, \dots, s_{t-4}$ and makes the conditional density depend only on the current regime s_t^* and not on past regimes.

A more general autoregressive model is developed by Lam (1990) as:

$$y_t - \mu_{s_t} = z_t - z_{t-1}, \quad \text{where } z_t = \phi_1 z_{t-1} + \phi_2 z_{t-2} + \dots + \phi_p z_{t-p} + \xi_t.$$

Although the whole history of the markov process is involved in this model, it turns out

that ξ_t depends only on $s_{t-p}, \dots, s_{t-1}, s_t$, and $\sum_{i=1}^t s_i$. Fortunately $\sum_{i=1}^t s_i$ is also Markovian and can be treated as an additional state variable. Hence he is able to transform the high-order Markov process into a first-order Markov process.

To allow time-varying state transition probabilities, Durland and McCurdy (1994) make state transitions depend on both the inferred current state j and also the number of periods the process has been in that state (d):

$$P\{s_t = j \mid \underbrace{s_{t-1} = j, \dots, s_{t-d} = j, s_{t-d-1} \neq j}_d\} = \exp(\alpha_j + \beta_j d) / (1 + \exp(\alpha_j + \beta_j d)), \text{ for } j = 1, 2.$$

This is a (d)-order Markov transition model. Since d is stochastic and can be very big, they merge all cases with $d > \tau$ ($\tau = 9$ in their case) as $d = \tau$ and hence they can transform the τ -order Markov process into a first-order Markov process with 2τ distinct states.

Both Cai (1994) and Hamilton and Susmel (1994) have extended the regime-switching in the mean process to describe the conditional variance by incorporating a low-order high-frequency ARCH process. Both argue that the regime-switching GARCH models are intractable due to the dependence of the conditional variance on the entire past history of the process. To deal with this problem, Gray (1996) provides a way of reducing computational complexity involved by approximating the path-dependent conditional variance of evolution of regimes with an inferred probability weighted path-independent conditional variance estimate. Hence the problem of path-dependence is solved in a way that preserves the essential nature of the GARCH process yet allows tractable estimation of the model.

2.3 Low-Frequency Path-Dependent Autoregressive Markov-Switching Model

This study models time-varying trend and volatility based on the low-frequency information over regime durations. The high-frequency dynamics is therefore not the focus of this paper. I will first define some concepts in low-frequency and then present a special model within the general framework of path-dependent model with regime-switching in (1) that has autoregressive process across regime durations.

I first characterize each realization of S_1^t by $s_t = j$ and three durations (d_0, d_1, d_2) if

$$(S_{t-d_0-d_1-d_2-1}, S_{t-d_0-d_1-d_2}, \dots, S_{t-d_0-d_1-1}, S_{t-d_0-d_1}, \dots, S_{t-d_0-1}, S_{t-d_0}, \dots, S_{t-1}, S_t = j) \\ = (\underbrace{L^3 j, L^2 j, \dots, L^2 j}_{d_2}, \underbrace{L j, \dots, L j}_{d_1}, \underbrace{j, \dots, j, j}_{d_0}),$$

where L is a lag operator or: $Lj = \begin{cases} j-1 & \text{if } j > 1 \\ J & \text{if } j = 1 \end{cases}$, $L^2 j = L(Lj)$, and $L^3 j = L(L^2 j)$.

I also assume that state switches in a non-skip fashion, or

$$P\{S_t = i \mid S_{t-1} = j, S_1^{t-2}, Y_1^{t-1}\} = \begin{cases} \geq 0 & \text{if } i = j \text{ or } i = Fj \\ = 0 & \text{o.w.} \end{cases}$$

where F is a forward operator or: $Fj = \begin{cases} j+1 & \text{if } j < J \\ 1 & \text{if } j = J \end{cases}$.

For example, $\{s_t=1, d_0=4, d_1=2, d_2=3\}$ can be used to represent one particular realization of S_1^t if $S_1^t = (s_1, s_2, \dots, s_t) = (\dots, 2, 1, 1, 1, 2, 2, 1, 1, 1, 1, 1)$ given $J=2$.

Only the most recent three durations are used in the model because these three durations together cover at least a full-cycle in the past given $J=2$. In addition, three durations in low-frequency in general cover a lot of lagged observations of a time series if it exhibits reasonably long duration like most economic time series do.

Further we aggregate corresponding Y_1^{t-1} information over each of the three past durations by their sample mean and standard deviation over each of the durations:

$$\tilde{\mu}_{0,S_t} = \frac{1}{d_0} \sum_{i=1}^{d_0} y_{t-i}, \quad \tilde{\mu}_{1,S_t} = \frac{1}{d_1} \sum_{i=1}^{d_1} y_{t-d_0-i}, \quad \text{and} \quad \tilde{\mu}_{2,S_t} = \frac{1}{d_2} \sum_{i=1}^{d_2} y_{t-d_0-d_1-i}.$$

The corresponding three standard deviation measures are: $\tilde{\sigma}_{0,S_t}$, $\tilde{\sigma}_{1,S_t}$, and $\tilde{\sigma}_{2,S_t}$.

Then one can specify equation (1) as a low-frequency model with low-order parsimonious mean and variance processes:

$$\mu(S_1^t, Y_1^{t-1}) = \mu(s_t, d_0, d_1, d_2, \tilde{\mu}_{0,S_t}, \tilde{\mu}_{1,S_t}, \tilde{\mu}_{2,S_t}), \quad \text{and} \quad \sigma(S_1^t, Y_1^{t-1}) = \sigma(s_t, d_0, d_1, d_2, \tilde{\sigma}_{0,S_t}, \tilde{\sigma}_{1,S_t}, \tilde{\sigma}_{2,S_t}).$$

Similar to Durland and McCurdy (1994), the state transitions can also depend on $(s_t, d_0, d_1, d_2, \tilde{\mu}_{0,S_t}, \tilde{\mu}_{1,S_t}, \tilde{\mu}_{2,S_t}, \tilde{\sigma}_{0,S_t}, \tilde{\sigma}_{1,S_t}, \tilde{\sigma}_{2,S_t})$.

Our low-frequency autoregressive model takes the following special form:

$$y_t = \beta_{s_t} + \beta_{0,S_t}(\tilde{\mu}_{0,S_t} - \beta_{s_t})d_0 + \beta_{1,S_t}(\tilde{\mu}_{1,S_t} - \beta_{L,S_t})d_1 + \beta_{2,S_t}(\tilde{\mu}_{2,S_t} - \beta_{L^2,S_t})d_2 + (\delta_{s_t} + \delta_{0,S_t}(\tilde{\sigma}_{0,S_t} - \delta_{s_t}) + \delta_{1,S_t}(\tilde{\sigma}_{1,S_t} - \delta_{L,S_t}) + \delta_{2,S_t}(\tilde{\sigma}_{2,S_t} - \delta_{L^2,S_t}))\xi_t \quad (2)$$

where ξ_t is i.i.d. $N(0,1)$. The transition probability is governed by:

$$P\{s_t = j \mid s_{t-1} = j, d_0, \tilde{\mu}_{0,S_t}\} = \Phi(\phi_{s_t} + \phi_{0,S_t}\tilde{\mu}_{0,S_t}d_0), \quad \text{and}$$

$$P\{s_t = F_j \mid s_{t-1} = j, d_0, \tilde{\mu}_{0,S_t}\} = 1 - P\{s_t = j \mid s_{t-1} = j, d_0, \tilde{\mu}_{0,S_t}\},$$

where $\Phi(\cdot)$ is the cumulative normal distribution function.

This model is presented in the error correction form around the long-term mean and variance. The Hamilton model relates $y_t - \beta_{s_t}$ with the previous deviation of y_{t-1} from their long term average $\beta_{s_{t-1}}$ for a given number of lagged observations in the high-frequency. This new model characterizes the current deviation of y_t from its long-term mean β_{s_t} in relating to differences between the sample average value of data and its long-term mean for each of most recent three durations. In the mean process of this model, the terms $\beta_{1,S_t}(\tilde{\mu}_{1,S_t} - \beta_{L,S_t})d_1$ and $\beta_{2,S_t}(\tilde{\mu}_{2,S_t} - \beta_{L^2,S_t})d_2$

$\beta_{t^2s_t})d_2$ measure the effect of cumulative deviations of the previous two durations' sample mean from their long-term mean of y_t . They impact in the exactly same way on the mean process for observations which include the current state and the number of periods the process has been in the state. Thus, this new model is able to generate time-varying trend across regime durations. The term $\beta_{0s_t}(\tilde{\mu}_{0s_t} - \beta_{s_t})d_0$ measures the cumulative deviation of the current duration's sample mean from their long-term mean of y_t up to time $t-1$. It also impacts all observations in the current duration but the impact varies depending on the cumulative deviation effect. The conditional variance in this model shares similar low-frequency properties as the mean process. Unlike Hamilton model, the number of lags in this new model varies depending on the stochastic values of d_0 , d_1 , and d_2 , making it difficult to transform the high-order Markov process into a simple first-order Markov process. In the next section, I will develop procedures to deal issues related to the path-dependency problem.

3. Maximum Likelihood Estimates

In this section, I will first provide the maximum likelihood estimation procedures for the general framework of path-dependent regime-switching model of equation (1). Then based on special properties of the low-frequency autoregressive specification of equation (2), a method is developed to reduce the computation intensity due to the path-dependency.

First I provide the maximum likelihood estimation procedures for the general framework of path-dependent regime-switching model of equation (1). Let θ be the set of parameters to be estimated. Then the conditional probability density function of y_t in equation (1) is :

$$\begin{aligned} & f(y_t | S_1^t, Y_1^{t-1}; \theta) \\ &= \{ \sqrt{2\pi} \sigma(S_1^t, Y_1^{t-1}; \theta) \}^{-1} \times \exp \{ - [y_t - \mu(S_1^t, Y_1^{t-1}; \theta)]^2 / [\sqrt{2} \sigma(S_1^t, Y_1^{t-1}; \theta)]^2 \} \end{aligned}$$

To calculate the log-likelihood value, one needs to iterate the following four equations with the guess of initial value of $P(S_1=j | y_0)$ for $j=1, \dots, J$. (θ is suppressed for the purpose of simplification of notation)

$$f(y_t, S_1^t | Y_1^{t-1}) = f(y_t | S_1^t, Y_1^{t-1}) P(S_1^t | Y_1^{t-1})$$

$$f(y_t | Y_1^{t-1}) = \sum_{S_1=1}^J \sum_{S_{t-1}=1}^J \dots \sum_{S_1=1}^J f(y_t, S_1^t | Y_1^{t-1})$$

$$P(S_1^t | Y_1^t) = f(y_t, S_1^t | Y_1^{t-1}) \div f(y_t | Y_1^{t-1})$$

$$P(S_1^{t+1} | Y_1^t) = P(s_{t+1} | S_1^t, Y_1^t) P(S_1^t | Y_1^t)$$

The initial value of $P(S_1=j | y_0) = p_j$ for $j=1, 2, \dots, J-1$ can be estimated as additional parameters in the model. The log-likelihood function is:

$$\ell(\theta) = \sum_{t=1}^T \log(f(y_t | Y_1^{t-1}; \theta)).$$

The filtered inferences about $s_t = j$ is :

$$P(s_t = j | Y_1^t) = \sum_{S_{t-1}=1}^J \sum_{S_{t-2}=1}^J \dots \sum_{S_1=1}^J P(s_t = j, S_1^{t-1} | Y_1^t).$$

The expected y_t given Y_1^{t-1} is :

$$E(y_t | Y_1^{t-1}) = \sum_{S_1=1}^J \sum_{S_{t-1}=1}^J \dots \sum_{S_1=1}^J E(y_t | S_1^t, Y_1^{t-1}) P(S_1^t | Y_1^{t-1})$$

Estimation of equation (1) requires consideration of all possible different realizations of whole past history of markov process. At time t , one needs to consider J^t different possibilities of S_1^t because each iteration of the Kalman filtering in the algorithm produces a J -fold increase in the number of past states to consider. Since this algorithm is too computationally burdensome, some approximation methods must be adopted to make the model tractable. One way to reduce the computation intensity is to restrict the number of past states to be considered

not to exceed τ such that if $t \leq \tau$, J^t possibilities of states need to be considered, and if $t > \tau$, only $J^{\tau+1}$ possibilities of states need to be considered.

The special characteristics of the low-frequency autoregressive model with Markov-switching present an opportunity for us to reduce the computation intensity involved in estimating model (1). Equation (2) asserts that at time t , only the information over $(t-d_0-d_1-d_2, \dots, t-1, t)$ is relevant and information beyond $t-d_0-d_1-d_2$ is not. So instead of considering all possibilities of S_1^t , we can alternatively consider all possible different values of (d_0, d_1, d_2) . Given $s_t = j$, the number of possibilities of (d_0, d_1, d_2) is:

$$\begin{aligned} \sum \text{of } & \left\{ \begin{array}{ll} (d_0, d_1, d_2) & \text{for } d_0 = t-1, d_1 = 0, d_2 = 0 \\ (d_0, d_1, d_2) & \text{for } d_0 + d_1 = t-1, d_0 \geq 0, d_1 \geq 1, d_2 = 0 \\ (d_0, d_1, d_2) & \text{for } d_0 + d_1 + d_2 \leq t-1, d_0 \geq 0, d_1 \geq 1, d_2 \geq 1 \end{array} \right. = \left. \begin{array}{l} 1 \\ t-1 \\ \sum_{d_0=0}^{t-3} \sum_{d_1=1}^{t-2-d_0} \sum_{d_2=1}^{t-1-d_0-d_1} 1 \end{array} \right\} \\ & = t + \frac{(t-2)(t-1)t}{6} \end{aligned}$$

In general, if n_d past durations are used, then one needs to consider the number of different realizations of past durations (n_d) in the order of $\frac{t!}{(t-n_d)!n_d!}$.

Similarly we can also put a constraint on the number of lagged states to be considered not to exceed τ so that if $t \leq \tau$, $J \cdot (t + \frac{(t-2)(t-1)t}{6})$ possibilities of markov process need to be considered, and if $t > \tau$, only $J \cdot (\tau+1 + \frac{(\tau-1)\tau(\tau+1)}{6})$ possibilities of Markov process need to be considered. For example, if $J = 2$ and $\tau = 20$, we only need to consider 2702 possibilities of markov process compared to $2 \times 2^{20} = 2.1 \times 10^6$ possibilities otherwise for $t > 20$. The transformation

from state-space representation to duration-space representation in describing the path-dependence allows us to achieve significant computation reduction.

Next I will provide the maximum likelihood estimation procedures for the low-frequency autoregressive model with regime-switching based on the duration-space representation of model

(2). Denote $\theta = (\beta_{s_t}, \beta_{0,s_t}, \beta_{1,s_t}, \beta_{2,s_t}, \delta_{s_t}, \delta_{0,s_t}, \delta_{1,s_t}, \delta_{2,s_t}, \phi_{s_t}, \phi_{0,s_t}; s_t=1,2,\dots,J)$ and suppress θ sometimes for the purpose of simplification of notation. Let $d = d_0 + d_1 + d_2$. For each $\{s_t=j, d_0, d_1, d_2\}$, the conditional probability density function of y_t is:

$$f(y_t | \{s_t=j, d_0, d_1, d_2\}, Y_{1-t-d}^{t-1}, \theta) = \{ \sqrt{2\pi} \sigma(s_t, d_0, d_1, d_2, \tilde{\sigma}_{0,s_t}, \tilde{\sigma}_{1,s_t}, \tilde{\sigma}_{2,s_t}) \}^{-1} \times \\ \exp \{ - [y_t - \mu(s_t, d_0, d_1, d_2, \tilde{\mu}_{0,s_t}, \tilde{\mu}_{1,s_t}, \tilde{\mu}_{2,s_t})]^2 / [\sqrt{2} \sigma(s_t, d_0, d_1, d_2, \tilde{\sigma}_{0,s_t}, \tilde{\sigma}_{1,s_t}, \tilde{\sigma}_{2,s_t})]^2 \}$$

We can obtain the log-likelihood by iterating the following four steps:

step 1 gets the joint density given Y_{1-t-d}^{t-1} for each $\{s_t=j, d_0, d_1, d_2\}$:

$$f(y_t, \{s_t=j, d_0, d_1, d_2\} | Y_{1-t-d}^{t-1}) = f(y_t | \{s_t=j, d_0, d_1, d_2\}, Y_{1-t-d}^{t-1}) P(\{s_t=j, d_0, d_1, d_2\} | Y_{1-t-d}^{t-1})$$

step 2 calculates the likelihood function at time t by summing up the results

from step 1 for all $\{s_t=j, d_0, d_1, d_2\}$

$$f(y_t | Y_{1-t-d}^{t-1}) = \sum_{\{s_t=j, d_0, d_1, d_2\}} f(y_t, \{s_t=j, d_0, d_1, d_2\} | Y_{1-t-d}^{t-1})$$

step 3 gets the filtered probability of $\{s_t=j, d_0, d_1, d_2\}$ given Y_{1-t-d}^t :

$$P(\{s_t=j, d_0, d_1, d_2\} | Y_{1-t-d}^t) = f(y_t, \{s_t=j, d_0, d_1, d_2\} | Y_{1-t-d}^{t-1}) \div f(y_t | Y_{1-t-d}^{t-1})$$

step 4 calculates the one period ahead probability transition based on $\{s_t=j, d_0, d_1, d_2\}$:

$$P(\{s_{t+1}, (d_0, d_1, d_2)'\} | Y_{1-t-d}^t) = P(s_{t+1} | \{s_t=j, d_0, d_1, d_2\}, Y_{1-t-d}^t) P(\{s_t=j, d_0, d_1, d_2\} | Y_{1-t-d}^t)$$

if $d < \tau$:

$$\text{for } s_{t+1}=s_t=j, \quad (d_0, d_1, d_2)' = (d_0+1, d_1, d_2)$$

$$\text{for } s_{t+1}=Fs_t=Fj, \quad (d_0, d_1, d_2)' = (0, d_0+1, d_1)$$

if $d = \tau$:

for $s_{t+1}=s_t=j$, and

$$\begin{cases} \text{if } d_0 = \tau, d_1 = 0, d_2 = 0 & \text{then } (d_0, d_1, d_2)' = (d_0, 0, 0) \\ \text{if } d_0 \geq 0, d_1 \geq 1, d_2 = 0 & \text{then } (d_0, d_1, d_2)' = (d_0 + 1, d_1 - 1, 0) \\ \text{if } d_0 \geq 0, d_1 \geq 1, d_2 \geq 1 & \text{then } (d_0, d_1, d_2)' = (d_0 + 1, d_1, d_2 - 1) \end{cases}$$

for $s_{t+1}=F s_t=Fj$, and

$$\begin{cases} \text{if } d_0 = \tau, d_1 = 0, d_2 = 0 & \text{then } (d_0, d_1, d_2)' = (0, d_0, 0) \\ \text{if } d_0 \geq 0, d_1 \geq 1, d_2 = 0 & \text{then } (d_0, d_1, d_2)' = (0, d_0 + 1, d_1 - 1) \\ \text{if } d_0 \geq 0, d_1 \geq 1, d_2 \geq 1 & \text{then } (d_0, d_1, d_2)' = (0, d_0 + 1, d_1) \end{cases}$$

Since different values of $\{s_t=j, d_0, d_1, d_2\}$ can be transited into the same value of $\{s_{t+1},$

$(d_0, d_1, d_2)'\}$ (see example below), we need to add together all probabilities of

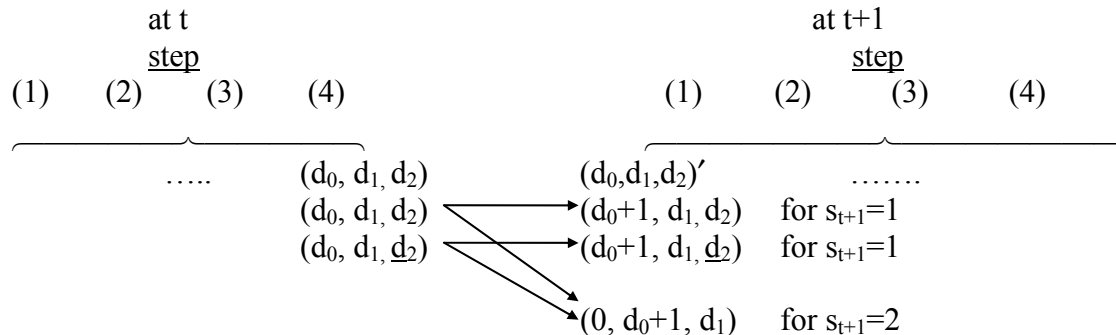
$P(\{s_{t+1}, (d_0, d_1, d_2)'\} | Y_{t-d}^t)$ associated with the same $\{s_{t+1}, (d_0, d_1, d_2)'\}$ and use the sum to

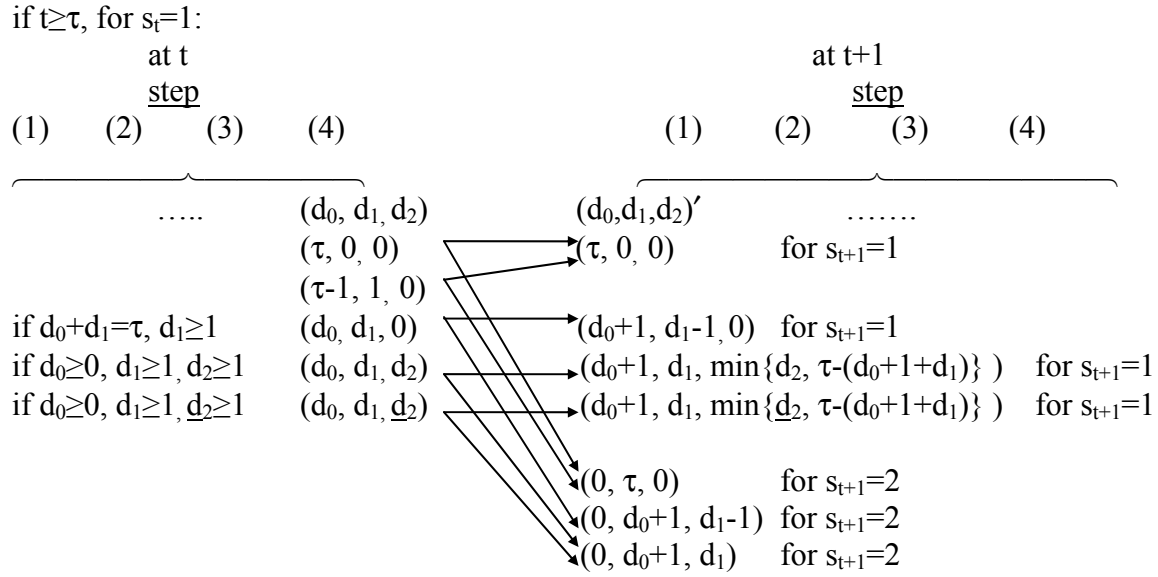
represent for $\{s_{t+1}, (d_0, d_1, d_2)'\}$.

Go back to step 1.

To facilitate the understanding of the above filter, I provide the following example to illustrate how $\{s_t=1, d_0, d_1, d_2\}$ evolves into $\{s_{t+1}, (d_0, d_1, d_2)'\}$ in step 4:

if $t < \tau$, for $s_t=1$:





The iteration from step 1 to step 4 can also generate some important by-products:

The log-likelihood function can be derived from step 2:

$$\ell(\boldsymbol{\theta}) = \sum_{t=1}^T \log (f(y_t | Y_1^{t-1}; \boldsymbol{\theta})).$$

The filtered inferences can be derived by summing up probabilities for all possible (d_0, d_1, d_2) associated with $s_t = j$ from step 3:

$$P(s_t = j | Y_1^t) = \sum_{(d_0, d_1, d_2)} P(\{s_t=j, d_0, d_1, d_2\} | Y_{t-d}^t)$$

The expected y_t given Y_1^{t-1} is the expectation of y_t probability-weighted over all

$\{s_t=j, d_0, d_1, d_2\}$ from step 1:

$$E(y_t | Y_1^{t-1}) = \sum_{\{s_t=j, d_0, d_1, d_2\}} E(y_t | \{s_t=j, d_0, d_1, d_2\}, Y_{t-d}^{t-1}) P(\{s_t=j, d_0, d_1, d_2\} | Y_{t-d}^{t-1})$$

We estimate the model by maximizing the log-likelihood function $\ell(\boldsymbol{\theta})$ using Berndt, Hall, Hall, and Hausman (1974) algorithm. The scoring is calculated numerically. The covariance matrix of the estimates is also computed numerically from the negative inverse of the Hessian of the log-likelihood function evaluated at these estimates.

4. An Empirical Example: The Behavior of Foreign Exchange Rates

The data are 100 times the first difference of the logarithmic exchange rates for four major currencies (British Pound, French Franc, German Mark, and Japanese Yen in foreign currency per U.S. dollar), beginning with the third quarter of 1973 and ending with the first quarter of 1994. Three models are compared using the data. These models include a linear (AR) specification, the original Hamilton model (1989), and the low-frequency autoregressive model with regime-switching. All models are in an error correction form:

Model 1: the single regime autoregressive model:

$$y_t - \beta_1 = \beta_{0,1} (y_{t-1} - \beta_1) + \beta_{1,1} (y_{t-2} - \beta_1) + \beta_{2,1} (y_{t-3} - \beta_1) + \beta_{3,1} (y_{t-4} - \beta_1) + \sigma \xi_t;$$

Model 2: the high-frequency autoregressive regime-switching model:

$$y_t - \beta_{s_t} = \beta_{0,s_t} (y_{t-1} - \beta_{s_{t-1}}) + \beta_{1,s_t} (y_{t-2} - \beta_{s_{t-2}}) + \beta_{2,s_t} (y_{t-3} - \beta_{s_{t-3}}) + \beta_{3,s_t} (y_{t-4} - \beta_{s_{t-4}}) + \delta_{s_t} \xi_t;$$

where s_t is presumed to follow a first order two-state Markov chain with transition probability:

$$P(s_t=1|s_{t-1}=1) = \Phi(\phi_1) \text{ and } P(s_t=2|s_{t-1}=2) = \Phi(\phi_2).$$

Model 3: the low-frequency autoregressive regime-switching model:

$$y_t - \beta_{s_t} = \beta_{0,s_t} (\tilde{\mu}_{0,s_t} - \beta_{s_t}) d_0 + \beta_{1,s_t} (\tilde{\mu}_{1,s_t} - \beta_{L,s_t}) d_1 + \beta_{2,s_t} (\tilde{\mu}_{2,s_t} - \beta_{L^2,s_t}) d_2 + \sigma_{s_t}(t) \xi_t;$$

where $\sigma_{s_t}(t) = \delta_{s_t} + \delta_{0,s_t} (\tilde{\sigma}_{0,s_t} - \delta_{s_t}) + \delta_{1,s_t} (\tilde{\sigma}_{1,s_t} - \delta_{L,s_t}) + \delta_{2,s_t} (\tilde{\sigma}_{2,s_t} - \delta_{L^2,s_t})$, with $(\tilde{\mu}_{j,s_t}, \tilde{\sigma}_{j,s_t})$ denoting y_t 's sample mean and std. deviation over the previous j -th duration, and s_t is presumed to follow a two-state Markov chain with transition probability:

$$P(s_t=1 | s_{t-1}=1, d_0, \tilde{\mu}_{0,1}) = \Phi(\phi_1 + \phi_{0,1} \tilde{\mu}_{0,1} d_0) \text{ and}$$

$$P(s_t=2 | s_{t-1}=2, d_0, \tilde{\mu}_{0,2}) = \Phi(\phi_2 + \phi_{0,2} \tilde{\mu}_{0,2} d_0).$$

According to the NBER's dating of business cycles, the average length of a business cycle is about 14 quarters. So $\tau=20$ quarters should be adequate on average to form the explanatory variables in the low-frequency regime-switching model.

Table 1 reports the maximum likelihood estimates for the three models. It shows that the low-frequency path-dependent regime-switching model with autoregressive mean and conditional variance achieves higher log-likelihood value than the linear AR(4) single-regime model and the Hamilton high-frequency regime-switching model for all four currencies.

Table I. Maximum Likelihood Estimates for Exchange Rates of British Pound, French Franc, German Mark, and Japanese Yen

The sample contains quarterly data from 1973Q3 to 1994Q1. The time series variable y_t is 100 times the first difference of log foreign exchange rate in foreign currency per U.S. dollar. Let ξ_t is i.i.d. $N(0,1)$. The following three models are estimated:

(1) Single regime autoregressive model:

$$y_t - \beta_1 = \beta_{0,1} (y_{t-1} - \beta_1) + \beta_{1,1} (y_{t-2} - \beta_1) + \beta_{2,1} (y_{t-3} - \beta_1) + \beta_{3,1} (y_{t-4} - \beta_1) + \delta_1 \xi_t;$$

(2) High-frequency autoregressive regime-switching model:

$$y_t - \beta_{s_t} = \beta_{0,s_t} (y_{t-1} - \beta_{s_{t-1}}) + \beta_{1,s_t} (y_{t-2} - \beta_{s_{t-2}}) + \beta_{2,s_t} (y_{t-3} - \beta_{s_{t-3}}) + \beta_{3,s_t} (y_{t-4} - \beta_{s_{t-4}}) + \delta_{s_t} \xi_t;$$

where s_t is presumed to follow a first order two-state Markov chain with transition probability:

$$P(s_t=1|s_{t-1}=1) = \Phi(\phi_1) \text{ and } P(s_t=2|s_{t-1}=2) = \Phi(\phi_2)$$

(3) Low-frequency autoregressive regime-switching model:

$$y_t - \beta_{s_t} = \beta_{0,s_t} (\tilde{\mu}_{0,s_t} - \beta_{s_t})d_0 + \beta_{1,s_t} (\tilde{\mu}_{1,s_t} - \beta_{L,s_t})d_1 + \beta_{2,s_t} (\tilde{\mu}_{2,s_t} - \beta_{L^2,s_t})d_2 + \sigma_{s_t}(t) \xi_t;$$

where $\sigma_{s_t}(t) = \delta_{s_t} + \delta_{0,s_t} (\tilde{\sigma}_{0,s_t} - \delta_{s_t}) + \delta_{1,s_t} (\tilde{\sigma}_{1,s_t} - \delta_{L,s_t}) + \delta_{2,s_t} (\tilde{\sigma}_{2,s_t} - \delta_{L^2,s_t})$, with $(\tilde{\mu}_{j,s_t}, \tilde{\sigma}_{j,s_t})$ denoting y_t 's mean and std. deviation over the previous j -th duration, and s_t is presumed to follow a two-state Markov chain with transition probability: $P(s_t=1|s_{t-1}=1) = \Phi(\phi_1 + \phi_{0,1} \tilde{\mu}_{0,1} d_0)$ and $P(s_t=2|s_{t-1}=2) = \Phi(\phi_2 + \phi_{0,2} \tilde{\mu}_{0,2} d_0)$.

Let $\beta_{s_t} = (\beta_{s_t}, \beta_{0,s_t}, \beta_{1,s_t}, \beta_{2,s_t})$, $\delta_{s_t} = (\delta_{s_t}, \delta_{0,s_t}, \delta_{1,s_t}, \delta_{2,s_t})$, and $\phi_{s_t} = (\phi_{s_t}, \phi_{0,s_t})$. Let ℓ to be log-likelihood value. The following table contains coefficients' MLE estimates, t-statistics(in parentheses), and the log-likelihood value:

(A) British Pound

(1) Single Regime Autoregressive Model		($\ell = -256.285$)			
Single-Regime	$\beta_1 =$	0.677(0.85)	0.305(2.31)	-0.217(-1.85)	0.231(2.18)
	$\delta_1 =$	5.305(11.78)			
(2) High-Frequency Autoregressive Regime-Switching Model		($\ell = -249.110$)			
Regime 1	$\beta_1 =$	4.751(7.15)	0.611(3.04)	-0.734(-4.12)	0.298(1.82)
	$\delta_1 =$	2.738(4.16)			
	$\phi_1 =$	0.575(1.87)			
Regime 2	$\beta_2 =$	-2.607(-3.23)	-0.190(-0.76)	-0.193(-0.97)	0.078(0.39)
	$\delta_2 =$	3.968(7.43)			
	$\phi_2 =$	0.747(2.41)			
(3) Low-Frequency Autoregressive Regime-Switching Model		($\ell = -245.266$)			
Regime 1	$\beta_1 =$	4.981(39.21)	-0.236(-2.09)	0.463(1.97)	-0.924(-3.99)
	$\delta_1 =$	2.239(2.13)	-0.069(-0.50)	0.676(1.57)	-0.268(-2.21)
	$\phi_1 =$	0.462(1.27)	0.001(0.00)		
Regime 2	$\beta_2 =$	-2.213(-17.9)	-0.484(-3.87)	0.636(9.04)	0.418(11.57)
	$\delta_2 =$	1.600(1.40)	0.436(2.34)	0.503(2.61)	-0.096(-0.72)
	$\phi_2 =$	0.366(1.14)	-0.012(-0.42)		

(B)French Franc

(1)Single Regime Autoregressive Model							(ℓ=-257.479)
Single-Regime	$\beta_1=$	0.372(0.40)	0.300(2.67)	-0.108(-0.92)	0.136(1.21)	0.011(0.09)	
	$\delta_1=$	5.382(9.85)					
(2)High-Frequency Autoregressive Regime-Switching Model							(ℓ=-253.435)
Regime 1	$\beta_1=$	4.128(7.86)	-0.267(-0.87)	-0.523(-1.71)	-0.315(-2.21)	-0.205(-0.96)	
	$\delta_1=$	3.807(5.69)					
	$\phi_1=$	0.911(2.84)					
Regime 2	$\beta_2=$	-3.426(-3.09)	0.142(0.67)	-0.032(-0.18)	0.135(0.72)	0.086(0.41)	
	$\delta_2=$	3.983(5.55)					
	$\phi_2=$	0.969(2.53)					
(3)Low-Frequency Autoregressive Regime-Switching Model							(ℓ=-240.737)
Regime 1	$\beta_1=$	4.471(38.97)	-0.548(-2.65)	0.132(2.51)	0.306(1.39)		
	$\delta_1=$	2.860(4.50)	-0.356(-2.43)	-0.404(-0.90)	-0.323(-1.03)		
	$\phi_1=$	0.697(1.91)	0.006(0.59)				
Regime 2	$\beta_2=$	-5.500(-7.12)	0.039(0.87)	-0.605(-2.93)	0.019(0.39)		
	$\delta_2=$	3.910(1.45)	0.275(0.85)	-0.379(-1.01)	0.244(0.38)		
	$\phi_2=$	1.565(3.34)	0.066(1.65)				

(C)German Mark

(1)Single Regime Autoregressive Model						(ℓ=-260.720)
Single-Regime	$\beta_1=$	-0.540(-0.62)	0.195(1.67)	-0.146(-1.21)	0.190(1.65)	0.046(0.39)
	$\delta_1=$	5.595(10.01)				
(2)High-Frequency Autoregressive Regime-Switching Model						(ℓ=-253.217)
Regime 1	$\beta_1=$	1.710(1.57)	0.395(1.55)	-0.315(-1.63)	0.442(1.83)	-0.202(-1.01)
	$\delta_1=$	4.462(6.91)				
	$\phi_1=$	0.654(2.26)				
Regime 2	$\beta_2=$	-6.938(-6.67)	0.516(4.40)	0.152(0.91)	-0.213(-1.00)	-0.103(-0.62)
	$\delta_2=$	1.533(2.03)				
	$\phi_2=$	-0.834(-1.85)				
(3)Low-Frequency Autoregressive Regime-Switching Model						(ℓ=-251.456)
Regime 1	$\beta_1=$	7.434(17.29)	0.386(0.76)	-0.107(-1.07)	0.198(0.89)	
	$\delta_1=$	2.390(0.95)	0.304(0.90)	0.353(0.80)	0.356(0.83)	
	$\phi_1=$	0.360(0.63) -0.089(-1.03)				
Regime 2	$\beta_2=$	-3.481(-8.30)	-0.183(-2.56)	0.871(3.76)	0.105(0.83)	
	$\delta_2=$	4.888(3.80)	0.271(0.85)	-0.212(-1.66)	-0.119(-0.19)	
	$\phi_2=$	0.907(2.71) -0.007(-0.42)				

(D)Japanese Yen

(1)Single Regime Autoregressive Model						(ℓ=-254.929)
Single-Regime	$\beta_1 =$	-1.099(-1.30)	0.246(2.07)	-0.062(-0.51)	0.151(1.43)	-0.052(-0.45)
	$\delta_1 =$	5.219(11.05)				
(2)High-Frequency Autoregressive Regime-Switching Model						(ℓ=-251.574)
Regime 1	$\beta_1 =$	3.662(5.07)	-0.107(-0.34)	-0.109(-0.48)	0.244(1.67)	-0.108(-0.51)
	$\delta_1 =$	2.562(4.05)				
	$\phi_1 =$	0.346(0.92)				
Regime 2	$\beta_2 =$	-3.864(-2.27)	0.242(1.27)	-0.112(-0.46)	0.260(1.10)	0.220(0.78)
	$\delta_2 =$	4.468(4.87)				
	$\phi_2 =$	0.953(2.30)				
(3)Low-Frequency Autoregressive Regime-Switching Model						(ℓ=-248.810)
Regime 1	$\beta_1 =$	1.683(1.82)	-0.131(-0.57)	-0.236(-0.89)	0.017(0.09)	
	$\delta_1 =$	3.708(3.21)	0.184(0.36)	0.165(0.42)	0.019(0.04)	
	$\phi_1 =$	0.791(1.53) -0.009(-0.19)				
Regime 2	$\beta_2 =$	-4.515(-3.49)	0.064(0.38)	0.241(1.21)	0.053(0.26)	
	$\delta_2 =$	4.043(5.12)	-0.398(-1.72)	-0.081(-0.22)	0.077(0.10)	
	$\phi_2 =$	0.702(0.93) -0.013(-0.44)				

Panel (1) of Table 1 reports estimates of the single-regime AR(4) model. Since parameters for two different regimes are not estimated separately, the long-term trend estimates for all currencies are not statistically significantly different from zero and the variance estimates tend to be relatively high when compared with the regime-switching models (in Panel (2) and (3)). Some reasonably significant coefficients for the short-term dynamics at $t-1$, $t-2$, $t-3$, $t-4$ are reported for all the currencies. It is worth noting that the coefficient corresponding to $t-1$ is always positive, $t-2$ is always negative, and $t-3$ is always positive, suggesting that all the four currencies move together. The relatively large magnitude of the coefficient corresponding to $t-1$ compared with the higher order lagged coefficients suggests that the all currencies tend to carry a strong momentum over time.

Panel (2) of Table 1 reports estimates of the high-frequency regime-switching model. Large difference in long-term trend coefficient exists between two regimes, indicating that changes of log foreign exchange rates are likely drawn from two different distributions. Except for regime 2 (U.S. dollar depreciation state) of German Mark, the persistence of regimes are reasonably high ($P\{s_t = j \mid s_{t-1} = j\} = \Phi(\phi_{st}) > \Phi(0.346) = 0.64$). Significant short-term dynamics are only found for British Pound's regime 1 (U.S. dollar appreciation state) and German Mark's regime 2. In both cases, strong positive coefficient corresponding to error correction term at $t-1$ suggests that there is a momentum within each duration, provided $s_{t-1} = s_t$ (which can be reasonably assumed because of the high persistence in regime transition).

Panel (3) of Table 1 reports estimates of the low-frequency regime-switching model. The long-term trend, volatility, and constant transition probability for each regimes are statistically significant. In the low-frequency regime-switching model, except for Japanese Yen, some significant low-frequency autoregressive processes are found. The coefficient of the cumulative

deviation from its long-term trend in current duration, if found to be significant, is always negative, suggesting a convergence toward the long-term trend. The impact of the cumulative deviation from the long-term trend in the two previous durations is not one-directional, though it is statistically significant sometimes. Except for British Pound's regime 2, there is no significant low-frequency variance process, suggesting volatility clustering mainly takes place in short run. Finally, we do not find evidence which indicates that the Markov transition probability is path-dependent for any of the currencies.

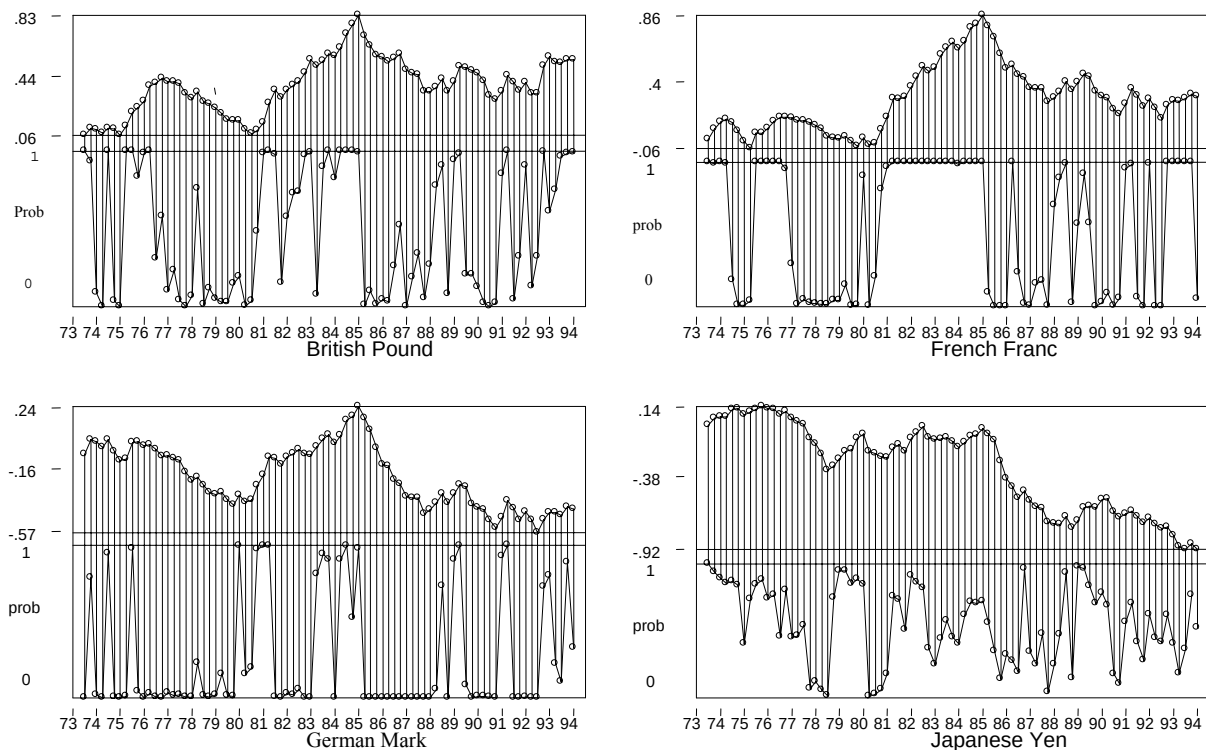


Figure 1. Log of Foreign Exchange Rates (Upper half) and the Filtered Probabilities $P(s_t = 1 | Y_1^t)$ (Lower half)
The foreign exchange rates (British Pound, French Franc, German Mark, and Japanese Yen) are in foreign currency per U.S. dollar normalized at time of 1973Q2 and are taken logarithm. The change in log foreign exchange rates is the dependent variable. The filtered probability that the change in log foreign exchange rate is in regime 1 (growth regime) at time t given all the information available at time t is derived based on the low-frequency regime-switching model. The sample contains quarterly data from 1973Q3 to 1994Q1.

Figure 1 shows the quarterly logarithm foreign exchange rates for British Pound, French Franc, German Mark, and Japanese Yen and the filtered probabilities that U.S. dollar is in appreciation state, or $P(s_t = 1 | Y_1^t)$. Except for Japanese Yen, unambiguous probabilistic inferences of each observation have been illustrated. For Japanese Yen, although the probabilistic inferences of most observation are not very clear-cut, they still tend to highly correlate with the directions of the movement of Japanese Yen.

5. Conclusion

In this paper, Hamilton's Markov-switching model has been extended to a low-frequency autoregressive process in the mean, variance, and markov transition probability in order to capture the time varying trend and conditional volatility of economic time series based on regime switching. Although this model is designed to capture the low-frequency (regime duration domain) time series dynamics, it does not exclude the usual high-frequency error correction terms.

Based on the state-space representation of regime-switching process, the estimation of the low-frequency autoregressive model is virtually intractable. This is because each iteration of the Kalman filtering in the algorithm produces a J -fold increase in the number of past states to consider, where J is the total number of states or regimes at each point in time. We reduce the computation intensity by transforming the state-space representation to duration-space representation in describing the Markov process and truncating the most recent few regime durations by a maximum allowable total duration. The algorithm to estimate the model by the method of maximum likelihood is also presented.

The maximum likelihood estimates, the basic filtering procedures presented in this paper produce results that are both statistically and economically significant for the quarterly data of foreign exchange rates over 1973-1994.

References:

- Bekaert, Geert and C. Harvey 1995 'Time-Varying World Market Integration'
Journal of Finance Vol 50: 403-44
- Berndt, E.K., B.H. Hall, R.E. Hall, and J.A. Hausman 1974 'Estimation and Inference in Non-linear Structure Models' Annals of Economic and Social Measurement Vol 4: 653-65
- Chen, Nai-fu 1991 'Financial Investment Opportunities and the Macroeconomy.'
Journal of Finance Vol 46: 529-54
- Chen, Nai-fu, R. Roll, and S. Ross 1986 'Economic Forces and the Stock Market.'
Journal of Business Vol 59: 383-403
- Cai, Jun 1994 'A Markov Model of Unconditional Variance in Arch' Journal of Business and Economic Statistics 12, 309-16
- Cochrane, John 1988 'How Big is the Random Walk in GDP.' Journal of Political Economy Vol 96: 893-920
- Daniel, K. and W. Torous 1991 'Common Stock Returns and the Business Cycle.'
UCLA working paper
- Diebold, F. X., J-H. Lee, and G.C. Weinbach 1994 'Regime Switching with Time-Varying Transition Probabilities' in C. Hargreaves, ed., Nonstationary Time Series Analysis and Cointegration (Oxford University Press, Oxford): 283-302
- Durland, J. Michael and T.H. McCurdy 1994 'Duration-Dependent Transitions in a Markov Model of U.S. GNP Growth' Journal of Business & Economic Statistics Vol 12 No.4: 279-88
- Engel, Charles and J. Hamilton 1990 'Long Swings in the Dollar: Are They in the Data and Do Markets know It?' American Economic Review Vol 80: 689-713
- Fama, E. 1990 'Stock Returns, Expected Returns, and Real Activities.'
Journal of Finance Vol XLV: 1089-108
- Fama, E. and K. French 1988 'Permanent and Temporary Components of Stock Prices.'
Journal of Political Economy Vol 96: 246-73
- Fama, E. and K. French 1989 'Business Condition and Expected Returns on Stocks and Bonds.' Journal of Financial Economics Vol 25: 23-49
- Gray, Stephen 1996 'Modeling the Conditional Distribution of Interest Rates as a

- Regime-Switching Process' *Journal of Financial Economics* 42: 27-62
- Goodwin, Thomas 1993 'Business Cycle Analysis with a Markov-switching Model.' *Journal of Business and Economic Statistics* 11:331-39
- Hamilton, James 1989 'A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle' *Econometrica* 57; 157-84
- Hamilton, James 1990 'Analysis of Time Series Subject to Changes in Regime.' *Journal of Econometrics* 45: 39-70
- Hamilton, James and R. Susmel 1994 'Autoregressive Conditional Heteroscedasticity and Changes in Regime' *Journal of Econometrics* 64: 307-33
- Kim, Chang-jin 1994 'Dynamic Linear Models with Markov-switching.' *Journal of Econometrics* 53: 1-22
- Lam, Pok-sang 1990 'The Hamilton Model with a General Autoregressive Component' *Journal of Monetary Economics* 26: 409-32
- Shumway, R.H. and D.S. Stoffer 1991 'Dynamic Linear Models with Switching.' *Journal of the American Statistical Association* Vol 86 No.415: 763-69
- Stock, James and M. Watson 1988 'Variable Trends in Economic Time Series.' *Journal of Economic Perspectives* Vol 2 No.3: 147-74